Computer Vision

Homework 0: Reading, Writing and Changing Images

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Based on Assignment 0 of CS 585, Image and Video Computing (M. Betke, Boston University, Spring 2024). Goes with Chapter 1 of the notes.

The point of this first homework is to get comfortable with images as arrays before anything clever happens. Use Python with NumPy and OpenCV (or a notebook), and take a photo of your own face, or of anything you like, to work with.

1. **Loading.** Read your photo with `cv2.imread`, check that it actually loaded (remember that a wrong file name gives `None`, not an error), and print its shape and data type. Which number in the shape is the width?

2. **Two kinds of grey.** Convert the photo to greyscale twice, once with the plain average $(R + G + B)/3$ written yourself and once with `cv2.cvtColor`. Be careful with the overflow of `uint8` arithmetic. Save both, show the absolute difference between them as a third image, and explain in a few sentences where in your photo the two disagree most and why.

3. **A creative transformation.** Design a second transformation of your own that produces an interesting result. One option is a blur in which every pixel is replaced by the average of itself and its 8 neighbours, written with array slicing rather than `cv2.blur`, and applied several times so that the effect is visible. Other ideas are welcome, as long as you write the core of it yourself.

4. **A deliberate mistake.** Reproduce one of the pitfalls of Sections 1.1.4 to 1.1.6 on purpose (a wrong width when reading raw bytes, swapped colour channels, or wrap-around) and include the broken picture with a short explanation of what you see.

Hand in one folder (or one notebook) with your code, the input photo and every output image. Say who you worked with and acknowledge any help, including help from AI tools.